Exam 1: Large Language Models

Problem 1

Consider a softmax model over categories, with a total of C possible categories. Assume that the probability of the observed category $y \in \{1, \ldots, C\}$ is

$$p(Y = y|F = f) = \frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)} \tag{1}$$

where $f \in \mathbb{R}^{d'}$, and each $w_c \in \mathbb{R}^{d'}$, for $c \in \{1, \ldots, C\}$. Assume that the set of vectors $\{w_c\}_{c=1,C}$ is known and fixed.

(a) Assume that we are given the observed category $y \in \{1, \ldots, C\}$, and our objective is to infer the associated latent vector $f \in \mathbb{R}^{d'}$. **Show** that the *cross entropy* (negative log-likelihood) for this observation $Y = y$, with f as the unknown parameter, may be expressed as

$$\ell(f) = -w_y^T f + \log \sum_{c=1}^{C} \exp(w_c^T f) \tag{2}$$

(b) Assume that we *initialize* f at vector f_0, and that we seek to update it with one step of gradient descent. **Show** that one step of gradient descent yields the update

$$f_1 = f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}] \tag{3}$$

where w_y is the vector corresponding to observed category $y \in \{1, \ldots, C\}$, and

$$\mathbb{E}(w)_{|f_0} = \sum_{j=1}^{C} w_j \left[\frac{\exp(w_j^T f_0)}{\sum_{c=1}^{C} \exp(w_c^T f_0)} \right] \tag{4}$$

The learning rate is $\alpha > 0$. **Hint:** Calculate the gradient of $\ell(f)$ and evaluate it at $f = f_0$.

(c) Consider the model in Figure 1. A feedforward network $g_\theta(f_0, w_y)$ operates on the concatenated input (f_0, w_y), where f_0 is the initialized vector for f and w_y is the vector associated with observed category y. The output of $g_\theta(f_0, w_y)$ is $(f_{out}, 0_{d'})$, where $0_{d'}$ is a d'-dimensional zero vector. The skip connection yields the updated output

$$(f_0, w_y) \to (f_0 + f_{out}, w_y) \tag{5}$$

The final vector $f = f_0 + f_{out}$ is input to the softmax function. Our goal is to learn the model parameters θ such that the softmax output assigns high probability to the observed category y.

Show that the negative log-likelihood for θ is approximately

$$\ell(\theta) \approx -\log p(Y = y|f_0) - f_{out} \cdot [\nabla_f \log p(Y = y|f)]_{f=f_0} \tag{6}$$

where $f_{out} \cdot [\nabla_f \log p(Y = y|f)]_{f=f_0}$ denotes the inner product between f_{out} and the gradient of the log probability evaluated at $f = f_0$. **Hint:** Use the first-order Taylor expansion around $f = f_0$.

(d) Based on the above result, show that minimizing the negative log-likelihood requires the vector f_{out} to align with the gradient $[\nabla_f \log p(Y = y|f)]_{f=f_0}$. **Explain** why this alignment corresponds to one step of gradient descent. Conclude that

$$f_{out} = \alpha[w_y - \mathbb{E}(w)_{|f=f_0}] \tag{7}$$

and the updated output f after the skip connection is approximately

$$f_0 + f_{out} \approx f_0 + \alpha[w_y - \mathbb{E}(w)_{|f=f_0}] \tag{8}$$

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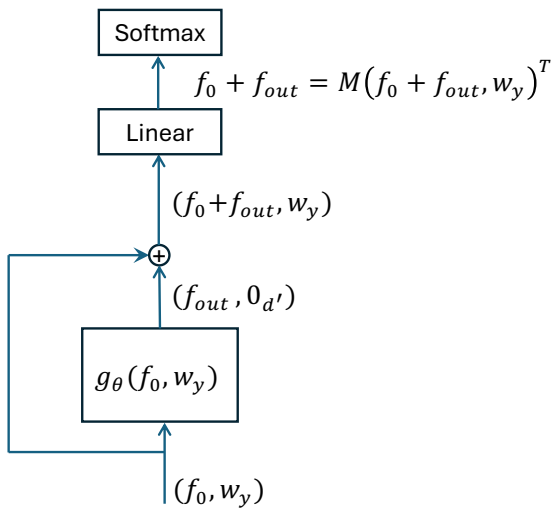

Figure 1: Skip connection model with a feedforward element $g_\theta(f_0, w_y)$ producing f_{out}. The final output is $f = f_0 + f_{out}$.

Problem 2

Consider contextual data $\{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{1, \ldots, C\}$. We are also given a query $x_{N+1} \in \mathbb{R}^d$, for which we wish to infer the associated category y_{N+1}. Assume that there is an underlying function $f(x) \in \mathbb{R}^{d'}$ such that

$$p(Y = y|f(x)) = \frac{\exp(w_y^T f(x))}{\sum_{c=1}^C \exp(w_c^T f(x))} \tag{9}$$

where $\{w_c\}_{c=1}^C$ are fixed and known.

Assume that $f(x) = A\phi(x)$, where $\phi(x) \in \mathbb{R}^n$ is a fixed nonlinear transformation of x to an n-dimensional feature space, and $A \in \mathbb{R}^{d' \times n}$ is an unknown, context-dependent matrix. Let a_m denote the mth row of A. Define the kernel $\kappa(x_i, x_j) = \phi(x_i)^T \phi(x_j)$.

(a) **Derive** the negative log-likelihood for the unknown matrix A using the contextual data $\{(x_i, y_i)\}_{i=1}^N$ and the softmax model.

(b) **Show** that one gradient descent update for the row a_m is given by

$$a_m^{(1)} = a_m^{(0)} + \alpha \sum_{i=1}^N [w_{y_i}(m) - \mathbb{E}[w(m)]|_{f^{(0)}(x_i)}]\phi(x_i) \tag{10}$$

where $w_{y_i}(m)$ is the mth component of w_{y_i}, and

$$\mathbb{E}[w(m)]|_{f^{(0)}(x_i)} = \sum_{c=1}^C w_c(m) \frac{\exp(w_c^T f^{(0)}(x_i))}{\sum_{c'=1}^C \exp(w_{c'}^T f^{(0)}(x_i))} \tag{11}$$

Hint: Use the chain rule $\nabla_{a_m} \ell(a_m) = [\nabla_{a_m} f_m(x)] \frac{\partial \ell[f(x)]}{\partial f_m(x)}$.

(c) Using the result of (b), **show** that the function $f(x_j)$ updates with gradient descent as

$$f^{(1)}(x_j) = f^{(0)}(x_j) + \alpha \sum_{i=1}^{N} [w_{y_i} - \mathbb{E}[w]_{|_{f^{(0)}(x_i)}}] \kappa(x_i, x_j) \tag{12}$$

(d) If the function is initialized as $f^{(0)}(x) = 0_{d'}$ for all x, **show** that

$$\mathbb{E}[w]_{|_{f^{(0)}(x_i)}} = \frac{1}{C} \sum_{c=1}^{C} w_c \tag{13}$$

Solutions (produced in LaTex format by GPT4o)

Problem 1

(a) **Solution:** We are asked to show that the cross-entropy for the observation $Y = y$ can be expressed as

$$\ell(f) = -w_y^T f + \log \sum_{c=1}^{C} \exp(w_c^T f).$$

Starting with the probability definition:

$$p(Y = y|F = f) = \frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)}.$$

The negative log-likelihood (cross-entropy) for the observed category y is defined as:

$$\ell(f) = -\log p(Y = y|F = f).$$

Substituting $p(Y = y|F = f)$:

$$\ell(f) = -\log \left(\frac{\exp(w_y^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)} \right).$$

Expanding the logarithm:

$$\ell(f) = -w_y^T f + \log \left(\sum_{c=1}^{C} \exp(w_c^T f) \right).$$

Hence, the desired form is shown.

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(b) **Solution:** We are tasked with showing that one step of gradient descent yields the update:

$$f_1 = f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}],$$

where

$$\mathbb{E}(w)_{|f_0} = \sum_{j=1}^{C} w_j \left[\frac{\exp(w_j^T f_0)}{\sum_{c=1}^{C} \exp(w_c^T f_0)} \right].$$

The gradient of the loss $\ell(f)$ is:

$$\nabla_f \ell(f) = -w_y + \sum_{j=1}^{C} w_j \left[\frac{\exp(w_j^T f)}{\sum_{c=1}^{C} \exp(w_c^T f)} \right].$$

Evaluating this at $f = f_0$:

$$\nabla_f \ell(f)\big|_{f=f_0} = -w_y + \mathbb{E}(w)_{|f_0}.$$

The gradient descent update rule is:

$$f_1 = f_0 - \alpha \nabla_f \ell(f)\big|_{f=f_0}.$$

Substituting the gradient:

$$f_1 = f_0 - \alpha(-w_y + \mathbb{E}(w)_{|f_0}).$$

Simplifying:

$$f_1 = f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}].$$

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(c) **Solution:** Given the skip connection model in Figure 1, the negative log-likelihood for the model parameters θ is:

$$\ell(\theta) = -\log p(Y = y|f_0 + f_{\text{out}}).$$

Using a Taylor expansion around $f = f_0$:

$$\log p(Y = y|f_0 + f_{\text{out}}) \approx \log p(Y = y|f_0) + f_{\text{out}} \cdot \nabla_f \log p(Y = y|f)\big|_{f=f_0}.$$

Thus:

$$\ell(\theta) \approx -\log p(Y = y|f_0) - f_{\text{out}} \cdot \nabla_f \log p(Y = y|f)\big|_{f=f_0}.$$

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(d) **Solution:** To minimize $\ell(\theta)$, we need to maximize the inner product $f_{\text{out}} \cdot \nabla_f \log p(Y = y|f)\big|_{f=f_0}$. The gradient of the log-probability at $f = f_0$ is:

$$\nabla_f \log p(Y = y|f)\big|_{f=f_0} = w_y - \mathbb{E}(w)_{|f_0}.$$

Thus, to maximize the inner product, we ideally set:

$$f_{\text{out}} = \alpha[w_y - \mathbb{E}(w)_{|f_0}],$$

where $\alpha > 0$.

With the skip connection, the output becomes:

$$f_0 + f_{\text{out}} \approx f_0 + \alpha[w_y - \mathbb{E}(w)_{|f_0}],$$

which corresponds to one step of gradient descent.

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Problem 2

(a) **Solution:** The negative log-likelihood for the data $\{(x_i, y_i)\}_{i=1}^N$ is:

$$\ell(A) = -\sum_{i=1}^{N} \log p(Y = y_i|f(x_i)),$$

where $f(x) = A\phi(x)$ and:

$$p(Y = y|f(x)) = \frac{\exp(w_y^T f(x))}{\sum_{c=1}^{C} \exp(w_c^T f(x))}.$$

Substituting $f(x_i) = A\phi(x_i)$:

$$\ell(A) = -\sum_{i=1}^{N} \log \left(\frac{\exp(w_{y_i}^T A\phi(x_i))}{\sum_{c=1}^{C} \exp(w_c^T A\phi(x_i))} \right).$$

Expanding the logarithm:

$$\ell(A) = -\sum_{i=1}^{N} \left[w_{y_i}^T A\phi(x_i) - \log \left(\sum_{c=1}^{C} \exp(w_c^T A\phi(x_i)) \right) \right].$$

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(b) **Solution:** Using the chain rule:

$$\nabla_{a_m} \ell(a_m) = \sum_{i=1}^{N} \left[\frac{\partial \ell(f(x_i))}{\partial f_m(x_i)} \cdot \nabla_{a_m} f_m(x_i) \right].$$

The gradient of $f_m(x_i) = a_m^T \phi(x_i)$ with respect to a_m is $\phi(x_i)$.
Thus:

$$\nabla_{a_m} \ell(a_m) = \sum_{i=1}^{N} \left(\frac{\partial \ell(f(x_i))}{\partial f_m(x_i)} \right) \phi(x_i).$$

The partial derivative $\frac{\partial \ell(f(x_i))}{\partial f_m(x_i)}$ is:

$$\frac{\partial \ell(f(x_i))}{\partial f_m(x_i)} = w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f(x_i)}}.$$

Hence:

$$\nabla_{a_m} \ell(a_m) = \sum_{i=1}^{N} \left[w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f(x_i)}} \right] \phi(x_i).$$

The gradient descent update is:

$$a_m^{(1)} = a_m^{(0)} + \alpha \sum_{i=1}^{N} \left[w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i).$$

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(c) **Solution:** We are tasked with deriving the updated function $f_m^{(1)}(x_j)$ after one step of gradient descent. Starting from the update rule for a_m:

$$a_m^{(1)} = a_m^{(0)} + \alpha \sum_{i=1}^{N} \left[w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i).$$

The function $f_m(x_j)$ is defined as $f_m(x_j) = a_m^T \phi(x_j)$. Substituting the updated $a_m^{(1)}$:

$$f_m^{(1)}(x_j) = \left(a_m^{(0)} + \alpha \sum_{i=1}^{N} \left[w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i) \right)^T \phi(x_j).$$

Expanding this expression:

$$f_m^{(1)}(x_j) = f_m^{(0)}(x_j) + \alpha \sum_{i=1}^{N} \left[w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \phi(x_i)^T \phi(x_j).$$

Using the kernel definition $\kappa(x_i, x_j) = \phi(x_i)^T \phi(x_j)$, we have:

$$f_m^{(1)}(x_j) = f_m^{(0)}(x_j) + \alpha \sum_{i=1}^{N} \left[w_{y_i}(m) - \mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} \right] \kappa(x_i, x_j).$$

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(d) **Solution:** When $f^{(0)}(x) = 0$, the softmax probabilities for all classes are equal:

$$p(Y = c | f^{(0)}(x_i)) = \frac{\exp(0)}{\sum_{c'=1}^{C} \exp(0)} = \frac{1}{C}.$$

The expected value $\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}}$ is given by:

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \sum_{c=1}^{C} w_c(m) \cdot p(Y = c | f^{(0)}(x_i)).$$

Substituting $p(Y = c|f^{(0)}(x_i)) = \frac{1}{C}$:

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \sum_{c=1}^{C} w_c(m) \cdot \frac{1}{C} = \frac{1}{C} \sum_{c=1}^{C} w_c(m).$$

Thus, the expected value simplifies to the average of the weights over all classes:

$$\mathbb{E}[w(m)]_{|_{f^{(0)}(x_i)}} = \frac{1}{C} \sum_{c=1}^{C} w_c(m).$$

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Food for thought

Note that for gradient-descent step $k \geq 0$ the functional update is

$$f^{(k+1)}(x_j) = f^{(k)}(x_j) + \alpha \sum_{i=1}^{N} [w_{y_i} - \mathbb{E}(w)_{|_{f^{(k)}(x_i)}}]\kappa(x_i, x_j)$$

Note that the text in blue is a *nonlinear* function of the latent function $f^{(k)}(x_i)$, via the expectation $\mathbb{E}(w)_{|_{f^{(k)}(x_i)}}$. This significantly complicates gradient descent inference for categorical observations. In the prior HW we consider real-valued outcomes, for which this issue doesn't arise.

Note from the first problem above, ideally (to first order) the feedforward element with skip connection calculates exactly what is in the blue font. This suggests that attention based on kernel $\kappa(x_i, x_j)$, along with a feedforward element and associated skip connection, can jointly implement the update of $f^{(k+1)}(x_j)$. This provides insight into the role of the feedforward element and skip connection in the Transformer, which *is* applied to categorical observations (tokens in language). This will be discussed further in class.

Further, coming back to 2(d), let's see what happens if the average embedding vector across the C categories is zero, i.e., $\frac{1}{C}\sum_{c=1}^{C} w_c = 0_{d'}$. Then we have from 2(c) that

$$f^{(1)}(x_j) = \alpha \sum_{i=1}^{N} w_{y_i} \kappa(x_i, x_j) \tag{14}$$

assuming we initialize all rows of A as zero vectors. This looks very much like the first attention layer of the Transformer, where the token-dependent embedding vectors are used as values. We would need to think about how to define x_i for language. If we define x_i as the embedding vector for the token plus the positional embedding vector, this is essentially the first attention layer of the Transformer (at least when viewed from the perspective of a single attention head).